PROJECT CRASHING

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Project Crashing

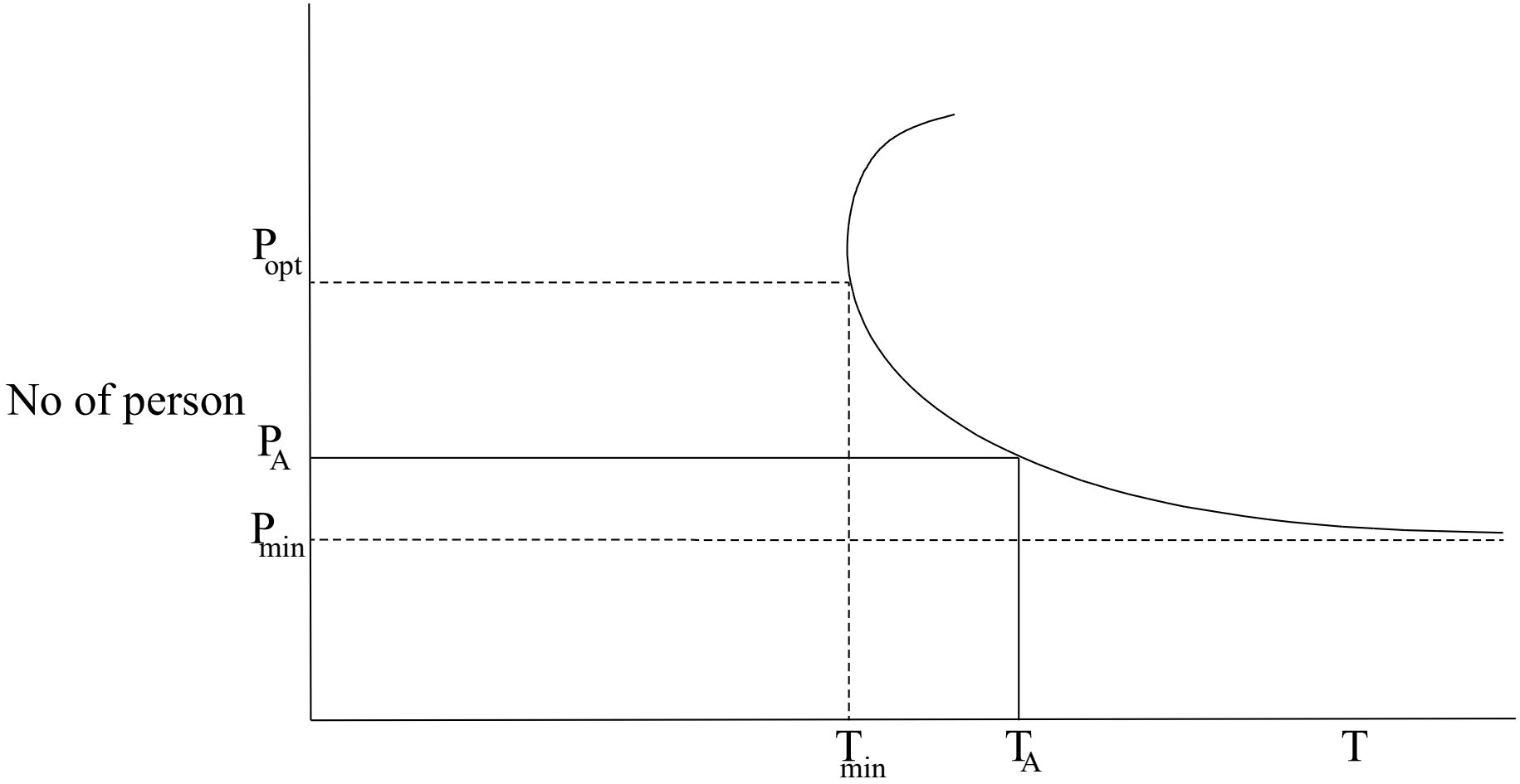
- While managing a project, a manager may face either (or both) of the following situations:
  - (1) the project is behind schedule
  - (2) the scheduled project completion time has been moved forward.

- In either situation, some or all of the remaining activities need to be speeded up (by adding resources) to finish the project by the desired due date.
- The process by which we shorten the duration of a project in the cheapest manner possible is called **project crashing**.

Crashing the project

- If after baselining the scheduled duration of critical path activities, the planned completion of the project misses the due date set by the customer, an alternative plan must be put in place to meet the market entry requirement.

- One technique that can be employed to shorten or crash the project is to increase the number of personnel in selected activities.
- Unfortunately, adding up to twice the personnel will not halve the duration. There are an optimum number of personnel who can be assigned to complete an activity.

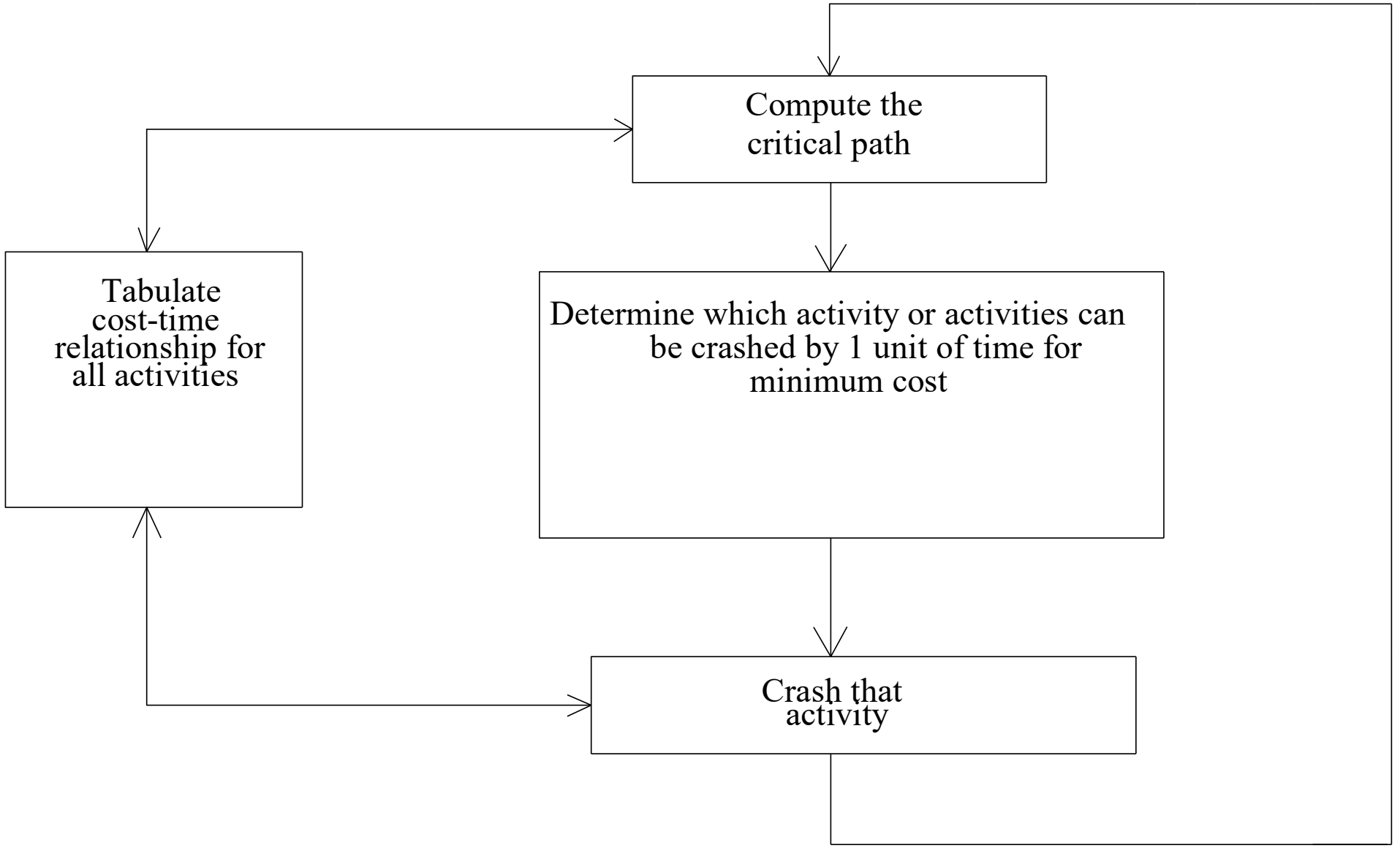

No of person
P_{opt}
P_A
P_{min}
T_{min}
T_A
T
Project completion time

- If an activity can be partitioned, project management can shorten the activity by increasing the number of appropriate personnel.
- However, increasing personnel often increases the need to communicate across a greater number of specialists, which may increase the actual time spent accomplishing the task.
- If the activity cannot be partitioned, additional personnel will not decrease the time of task.
- Suppose that two expert chefs are engaged in creating a delicious dish, how would the addition of a third reduce a customer's wait time for the dessert? Might third chef only increase conversation and conflict in the kitchen!

Which tasks to crash ?

1. Crash an activity only if it is on the critical path.
2. To minimize cost, crash the activity that costs the least to shorten. For instance, if two activities, A and B, are on the critical path, and if adding one person to activity A will decrease its duration by 2 months, but adding one person to activity B will decrease B by 3 months, add the one person to activity B.

Flow diagram for crashing a project



Example

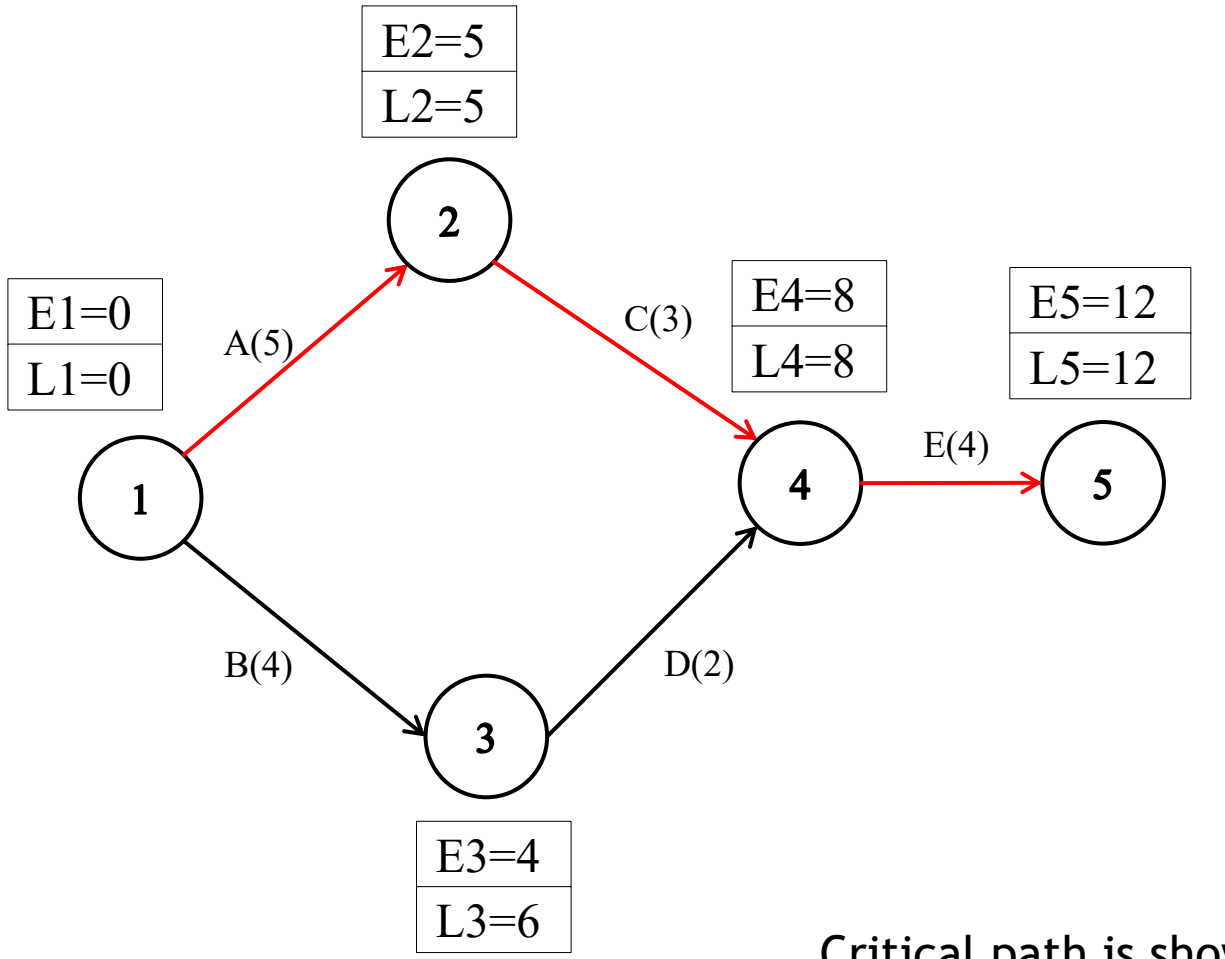
Activity	Description	Duration (days)	Predecessor	Crash cost/day (Rs)	Min duration (days)
A	Design engine	5	-	400	3
B	Build chassis	4	-	300	3
C	Assemble engine	3	A	500	2
D	Install wiring	2	B	200	1
E	Final assembly	4	C, D	600	2

A car manufacturing company wants to reduce the project duration to 10 days. Which activities should be crashed, and what will be the total additional (crashing) cost?

❖ Determine the critical path

E: Earliest occurrence time of an event

L: Latest occurrence time of an event



Critical path is shown in red colour.

The critical path of the project is: **1-2-4-5** and critical activities are **A**, **C**, and **E**.

Total project duration = sum of duration of critical activities

= (5 + 3 + 8) = 12 days.

Objective: Reduce total project duration from **12 days to 10 days.**

We must **crash 2 days** on the **critical path** only.

Minimum duration is the shortest duration a task can be completed, no matter how many resources are added to speed up the task.

Thus, activity A can be crashed from **5→3 days** (2 days max), activity C can be crashed from **3→2 days** (1 day max), and activity E can be crashed from **4→2 days** (2 days max).

Critical activities

Activity	Can be Crashed?	Crash cost/Day (Rs)	Days to Crash (days)
A	Yes (5 → 3)	400	2
C	Yes (3 → 2)	500	1
E	Yes (4 → 2)	600	2

Start with **lowest crash cost per day** on critical path:

Option 1

Crash activity A by 1 day → Rs. 400

and

Crash activity C by 1 day → Rs. 500

Total crashing cost = 400 + 500 = Rs. 900.

Option 2 ✅

Crash activity A by 2 days → Rs. 400×2 = Rs. 800

Total crashing cost = Rs. 800.

Both achieve the goal (10-days project)

Crashing **A by 2 days** is cheaper.

Total duration = 12 − 2 = 10 days,

Total crashing cost = 400×2 = Rs. 800.

Summary

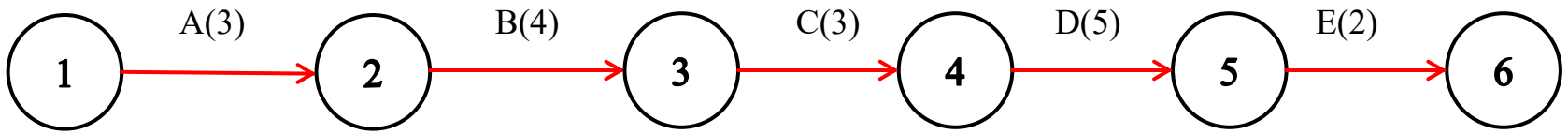
- Original duration: 12 days
- New duration: 10 days
- Crashing cost: Rs. 800
- Activities crashed: A (2 days).

Example

Activity	Description	Duration (days)	Predecessor	Crash cost/day (Rs)	Minimum duration (days)
A	Remove old furniture	3	-	100	2
B	Repair electrical wiring	4	A	200	2
C	Paint walls	3	B	250	2
D	Install new flooring	5	C	300	3
E	Set up new furniture	2	D	0	2

A renovation company wants to reduce the project duration to 14 days. Which activities should be crashed, and what will be the total additional (crashing) cost?

Determine the critical path



Critical path is shown in red colour.

The critical path of the project is: **1-2-4-5-6** and critical activities are **A**, **B**, **C**, **D**, and **E**.

Total project duration = sum of duration of critical activities

= (3+ 4 + 3 + 5 + 2) = 17 days.

Objective: Reduce total project duration from **17 days to 14 days.**

We must **crash 3 days** on the **critical path**.

Activity A can be crashed from **3→2 days** (1 day max), activity B can be crashed from **4→2 days** (2 days max), activity C can be crashed from **3→2 days** (1 day max), activity D can be crashed from **5→3 days** (2 days max), and activity E can be crashed from **2→2 days** (0 day max).

Critical activities

Activity	Can be Crashed?	Crash cost/Day (Rs)	Days to Crash (days)
A	Yes (3 → 2)	100	1
B	Yes (4 → 2)	200	2
C	Yes (3 → 2)	250	1
D	Yes (5 → 3)	300	2
E	No (2 → 2)	0	0

Start with **lowest crash cost per day** on critical path:

Crash activity A by 1 day → Rs. 100

and

Crash activity B by 2 days → Rs. 200×2 = Rs. 400

This crash combination achieves the target with the **least cost.**

Total duration = 17 − 3 = 14 days,

Total crashing cost = 100×1 + 200×2 = Rs. 500.

Summary

- Original duration: 17 days
- New duration: 14 days
- Crashing cost: Rs. 500
- Activities crashed: A (1 day) and B (2 days)